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Department of Computer Science and Engineering
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE
Course Code & Title: CCS339 Cryptocurrency and Blockchain Technologies
Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit V / Smart contracts
Course Outcome: CO 5

Unit Learning Outcome: UO 5a

Activity Chosen: Collaborative Learning

Justification:

- Collaborative learning, a dynamic teaching approach, fosters student engagement by organizing them into groups to collectively tackle challenges. Through this method, students pool their diverse perspectives and expertise to solve problems, enhancing their understanding of complex concepts while promoting intrinsic motivation.
- It also helps students make concepts more interesting by involving them to work together.

Time Allotted for the Activity: 40 minutes

Details of the Implementation:

- The students were divided into small groups, ensuring each group comprised 2 to 3 members.
- The task of preparing a conceptual understanding of smart contracts was assigned to each group.
- Group members were encouraged to conduct research independently and brainstorm ideas collectively.
- The groups collaboratively outlined the fundamental concepts and principles of smart contracts, including their definition, purpose, components, and potential applications.
- Each group was provided with an opportunity to present their conceptual understanding of smart contracts to the class.
- The activity concluded with a reflection session where students shared their insights, lessons learned, and areas for improvement.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PSO1
CO 5	2	2	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO5
	2	2	2	1	1
Justification for correlation	Students discussed the principles of blockchain technology and the role of smart contracts.	Students analyzed case studies of existing smart contract applications.	Students practiced writing and deploying smart contracts using Remix IDE.	Students analyzed and debugged smart contracts, addressing potential security vulnerabilities.	Students participated in practical sessions using Remix IDE for smart contract development and deployment.
	PO9	**PO10**	**PSO2**		
	1	1	1		
	Students discussed the topic as a member or leader together.	Students prepared and delivered presentations, about their smart contract.	Students will be able to apply blockchain concepts to develop reliable IT solutions.		

- **Images / Screenshot of the practice:**

Figure 1: Discussion with team members

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    o Students said that the activity was helpful in determining how well they understood the concept.

    o Students told the teacher that the activity encouraged them to ask more questions.

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students were able to explain the concepts clearly, which helped the instructor analyze, evaluate, and enhance their own teaching methods.
- This activity provided students with a deeper understanding of the specific topic.

❖ ***Challenges faced in implementation:***

- Most of the students actively participated except few students. They are not involved to share their understanding level and raising the doubts.
- Motivate the students those who are not involved in the activity effectively by means of addressing the benefits of collaborative learning.

References:

1. https://www.edsys.in/what-is-peer-teaching/
2. https://www.opencolleges.edu.au/informed/features/peer-teaching/
3. https://tilt.colostate.edu/TipsAndGuides/Tip/180
4. https://www.gdrc.org/kmgmt/c-learn/
5. https://www.edutopia.org/topic/collaborative-learning
6. https://www.ritrjpm.ac.in/images/computer-science/28.CS8501_ColloborativeLearning.pdf

30/4/24

Signature of Faculty Member

30/4/24

HOD